Enriching data

DATA201/422 Data Wrangling

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Coverage and representativeness

Extract from an email from a rental car company

As we are committed to improving our cars and services, we would appreciate your feedback on your overall impression of your experience. Please click one of the following two buttons to give us your feedback:

Satisfied with hire

Unsatisfied with hire

- Clicking on the left link led to a Google review page
- Clicking on the right link led to an internal company review page
- How do the two datasets differ?
- What would you miss if you only analysed the left-link database?
- What would you miss if you only analysed the right-link database?

Useful concepts for thinking about coverage

- **Target population**
  - Who/what are you trying to learn about?
- **Sample**
  - Who/what do you actually have information about?
- **Representativeness**
  - Do the units in your sample look like the population as a whole? (Allowing for random variation)
- **Selection bias**
  - If the units in your sample don’t look like those in your population, how does this affect your results?
- **Unit non-response**
  - You tried to sample a unit, but you obtained no information about it at all
  - Contrasts with **item non-response**, when you have partial info on the unit

Other examples of non-representative data

- Sellers typically only have data on people who bought their product
  - Not on people who almost bought it
  - (What does this assume about the target population?)
- Respondents to political polls
  - People who answer phone calls (5-10% of those approached in NZ)
- Fishing boats that allow observers

What to do if you have non-representative data

- Get data on the units who are missing
- If you can’t get more data (or if you can only get some data), then try statistical adjustments
  - Weighting
  - Post-stratification

Example of getting extra data: The 2018 NZ population census

Sources of enumeration and final census coverage, 2018 NZ Census

Ethnic group	Individual census form	Household listing only	Admin enumeration	Final coverage
Māori	68.0%	6.0%	21.7%	95.6%
Pacific peoples	64.5%	8.3%	22.4%	95.1%
Asian	81.2%	6.0%	9.4%	96.7%
European, other	87.0%	3.1%	8.1%	98.2%

Source: Stats NZ, *Post-enumeration survey 2018: Methods and results*, table 9.

Weighting

- Traditional approach to non-representative data
- Analysts construct weights which are added to analysis file

Weighting example

gender	number in sample	number in population	selection probability	weight
Female	40	80	0.5	2
Male	30	90	0.3333	3

id	gender	income (000)	weight
01	Female	78	2
02	Female	65	2
03	Male	57	3
04	Female	54	2
05	Male	92	3

Mean income for records 01 – 05:

(78 * 2 + 65 * 2 + 57 * 3 + 54 * 2 + 92 * 3) / (2 + 2 + 3 + 2 + 3) = 70.08

Problems with weighting

- To make the sample look more representative, you often have to adjust along multiple dimensions
  - Age, sex, geography, education, etc
- But weighting gets tricky (and sensitive) when you use lots of variables
- Little consensus on what to do

Post-stratification

- Alternative to weighting, becoming more popular
- Recipe
  1. Disaggregate sample to the point where you hope that representative within groups
  2. Fit model to disaggregated sample
  3. If necessary, use external data to aggregate results
- Step 2 is the hard part, but methods have been improving fast
- For more information, google “MrP”

Documentation

Metadata

- Metadata = data about data
  - Definitions of variables
  - Target population
  - Coverage of target population
  - How data were generated
  - etc
- Typically neglected
  - Harder than it looks
  - Few people enjoy constructing metadata
  - Major weakness of linked admin data systems like the IDI
- Essential part of data science pipeline
  - But incorrect metadata is even worse than no metadata

Description of data

- Inputs, cleaned data
- Means, frequencies, missing data cells, etc
- Graphs wherever possible
- Should be automated!!
  - Hand-crafted ones go out of date
  - Use Quarto, Jupyter notebooks, etc

Role of AI

Looking for data

- Finding data, concordances, metadata
- You are responsible for checking that the data matches the AI’s description

Writing the code

- AIs are good at this sort of code
- Write a detailed plan
  - Distinguish between “planning” mode and “agent” mode
- Design some tests
  - The agent can implement them

Automated descriptions of data

- Get the AI agent to make a summary of the inputs and outputs
  - In a Quarto/Jupyter notebook
- Data descriptions are a good task for AIs
  - Fairly automatic
  - Relatively easy to spot errors
  - Tedious and slow to do by hand
- Tell the agent to use R/Python
  - Allows easy repetition, extension
  - You can audit the important bits if you need to (ask the AI to help)

Things to watch out for

- Making up values
- Misunderstanding variables
- Sometimes descriptions of data emphasize weird things
- Overly complicated code